

Nonequilibrium Relaxation and the Odd-Even Effect in 2D Electron Gases

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Fermi Liquid Scaling

Introduction: The Fermi Liquid Scaling

- **Standard Fermi Liquid:** Decay rate of quasiparticles governed by $\gamma \sim T^2 / \hbar T_F$.
- **Odd-Parity Modes:** Predicted to have anomalously long lifetimes with $\gamma \sim T^4 / \hbar T_F^3$.
- **Goal:** Map the full spectrum of relaxational eigenmodes beyond the low- T asymptotic limit.

Kinetic Theory & Basis Expansion

- **Linearized Collision Integral:**

$$\hat{J}[\delta f] = -\frac{m^*}{4\pi\hbar^3} \int \dots F_{121'2'}[\psi_1 + \psi_2 - \psi_{1'} - \psi_{2'}] [7, 8].$$

- **Parametrization:** $\delta f = f_0(1 - f_0)\psi(t, p)$, where ψ is a momentum-dependent chemical potential variation [7].
- **Radial Basis Expansion:** $\psi_m(p) = \sum_{i=1}^N c_i u_i(w)$ using orthonormal polynomials derived from a Fermi-Dirac scalar product [9, 10].
- **Low- T Basis Functions:**
$$u_1 = 1, u_2 = \frac{\sqrt{3}}{\pi} w, u_3 = \frac{3\sqrt{5}}{4\pi^2} w^2 - \frac{\sqrt{5}}{4} [11].$$

Microscopic Origin: Even vs. Odd Parity

Even-Parity Modes Relax via **head-on scattering** ($P = 0$), which is efficient at high momentum transfer $O(k_F)$. Results in standard T^2 scaling

Lemma 1.1: *Odd-Parity Modes (Orange)* Cannot relax via head-on processes. They relax through repeated **small-angle scattering** (subdiffusive), leading to T^4 scaling [4, 14, 15].

- This subdiffusive behavior spreads localized deformations as $t^{1/4}$ [16].

The Spectrum of Decay Rates

- **Isolated Band:** There exists a distinct set of long-lived odd-parity modes separated by a large gap from the continuum [6, 17].
- **Mode Hierarchy:** The number of long-lived modes increases at low temperatures as $\sim \sqrt{T_F/T}$ [18, 19].
- **Continuum Scaling:** All higher-order eigenmodes in both sectors revert to standard T^2 Fermi-liquid behavior [6, 18].
- **Persistence:** The odd-even effect remains detectable up to $T \approx 0.15T_F$ [1, 20].

Tunability with Coulomb Interaction r_s

- **Interaction Strength (r_s):** Screened Coulomb potential

$$V(k) = \frac{2\pi e^2}{k + k_{TF}} [21, 22].$$

- **Even-Mode Sensitivity:** Strongly dependent on r_s because they involve momentum transfers of order $O(k_F)$ [13, 23].
- **Odd-Mode Independence:** Nearly independent of r_s due to overscreening at small angles [23, 24].
- **Experimental Control:** The gap between lifetimes is tunable via density (doping), which changes r_s [25].

Conclusion: The Tomographic Transport Regime

- **Beyond Hydrodynamics:** The large disparity in lifetimes allows for an intermediate regime: **Tomographic Transport** [26].
- **Timescales:** Exists for $\gamma_{even}^{-1} \ll t \ll \gamma_{odd}^{-1}$, where odd modes must be included in the dynamics [26, 27].
- **Summary:** We provide a numerically exact diagonalization that confirms anomalously long-lived modes in experimentally relevant regimes [1, 28].